EXPERIMENT - 12

AIM :

To implement a Tic Tac Toe game using Python for two players.

ALGORITHM:

1. Start the program and initialize a 3 × 3 board.
2. Display the board.
3. Get the position from the current player.
4. Check whether the position is valid and empty.
5. Place the player's symbol on the board.
6. Check for a win or a draw.
7. Repeat until the game ends.
8. Display the result and stop.

PROGRAM :

untitled (3.14.3)

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```
board = [" "] * 9

def display():
    print()
    for i in range(0, 9, 3):
        print(" " + " | ".join(board[i:i+3]))
        if i < 6:
            print("---+---+---")
    print()

def check_winner(player):
    wins = [
        (0, 1, 2), (3, 4, 5), (6, 7, 8),
        (0, 3, 6), (1, 4, 7), (2, 5, 8),
        (0, 4, 8), (2, 4, 6)
    ]
    for a, b, c in wins:
        if board[a] == board[b] == board[c] == player:
            return True
    return False

player = "X"

while True:
    display()
    try:
        move = int(input(f"Player {player}, enter position (1-9): ")) - 1
    except ValueError:
        print("Enter a number from 1 to 9.")
        continue

    if move < 0 or move > 8 or board[move] != " ":
        print("Invalid move! Try again.")
        continue

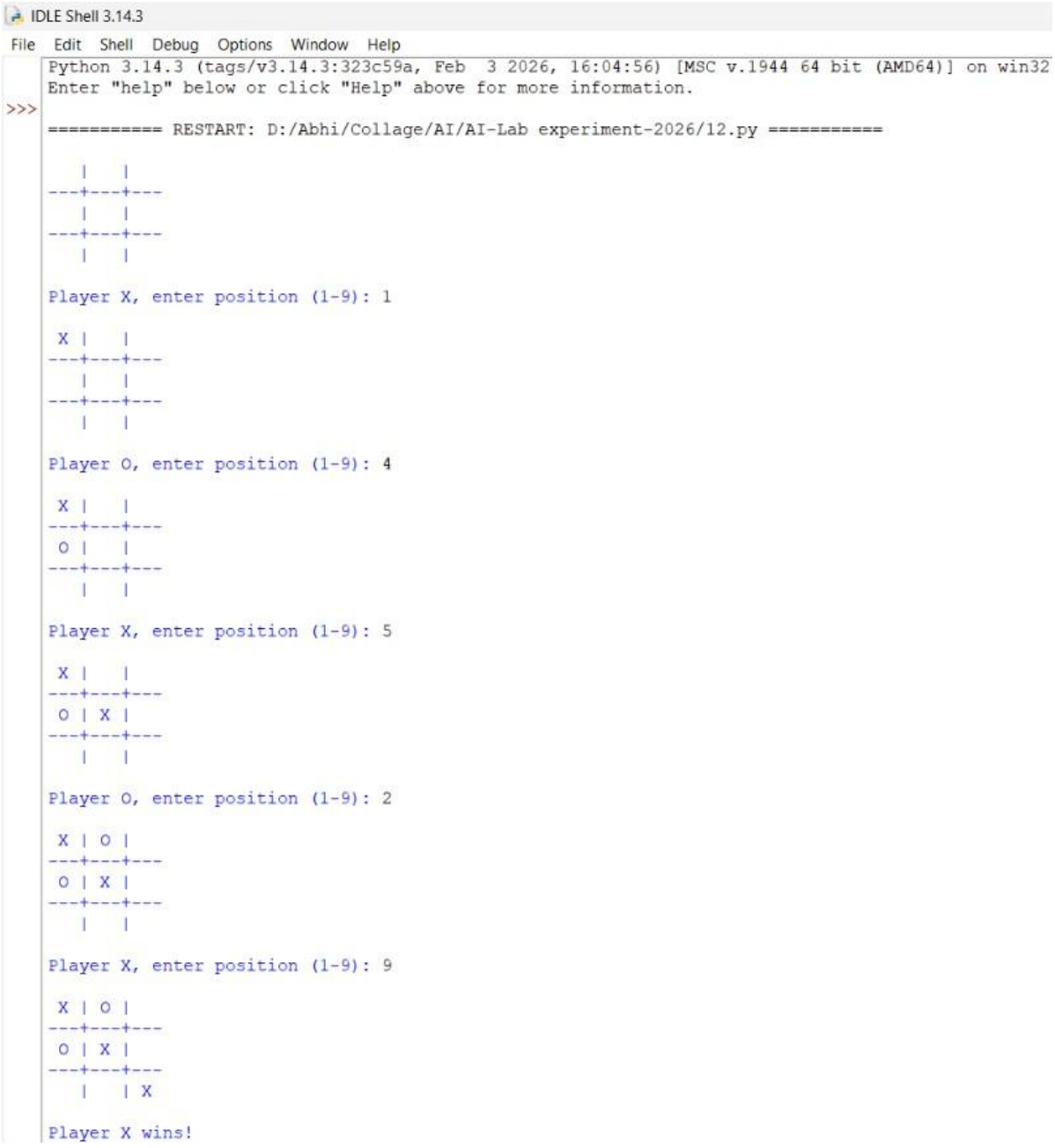
    board[move] = player

    if check_winner(player):
        display()
        print("Player", player, "wins!")
        break

    if " " not in board:
        display()
        print("Match Draw!")
        break

    player = "O" if player == "X" else "X"
```

OUTPUT:



RESULT:

Thus, the Tic Tac Toe game was implemented successfully, and the winner was identified.